VIJAY YADAV

PhD student working on machine learning surrogate models for soft matter physics.

Indian | ivijayyadav.github.io | yadav.59@iitj.ac.in | Publications | 0009-0006-4903-8044 | Jodhpur, IN

EDUCATION

PhD, Physics
Indian Institute of Technology Jodhpur — Rajasthan, India
12/2021 – Expected 12/2026

- Thesis title: Machine Learning Surrogate Models for Phase-Ordering Kinetics and Charge Regulation Effects.
- PhD advisor: Prabhat K. Jaiswal

Master of Science, Physics
National Institute of Technology Tiruchirappalli — Tamil Nadu, India
08/2018 – 06/2020

Bachelor of Science, Physics
University of Delhi — Delhi, India
07/2015 – 05/2018

PUBLICATIONS

Published

1. V. Yadav, P. K. Jaiswal, R. Podgornik, and S. Kumari, "Charge Regulation Effect on Nanoparticle Interaction Mediated by a Polyelectrolyte," *Macromolecules* **59**, **11**, 6398-6409 (2026). [Selected for Front Cover]
2. V. Yadav, M. Priya, M. D. Shrimali, and P. K. Jaiswal, "Graph Neural Network for Prediction of Phase-Ordering Kinetics," *Chaos* **35**, 061101 (2025).
3. S. Chauhan, S. Mandal, V. Yadav, P. K. Jaiswal, M. Priya, and M. D. Shrimali, "Machine Learning Based Prediction of Phase Ordering Dynamics," *Chaos* **33**, 061103 (2023).

Submitted

1. V. Yadav, M. Priya, M. D. Shrimali, and P. K. Jaiswal, "Physics-Guided Convolutional Neural Network for Domain Growth Prediction in Systems with Conserved Kinetics," manuscript submitted (2026). [arXiv:2606.26128]

In Preparation

1. V. Yadav, S. Kumari, and P. K. Jaiswal, "Machine Learning Accelerated Simulation of Charge Regulation Effects," (2026).

TEACHING EXPERIENCE

2024–26: Delivered lectures and tutorials on Python programming fundamentals, NumPy, and data visualization for M.Sc. third-semester students (PHP 6070) at IIT Jodhpur, India.
2023–25: Delivered lectures and tutorials on Python programming fundamentals and algorithms for B.S. third-semester students (PHP 2010) at IIT Jodhpur, India.

PRESENTATIONS

07/2026: Flash talk at the Symposium on Complex Dynamics in Chemistry, Physics, and Biology (CDCPB 2026) on "Data-Driven Modelling of Phase-Ordering Kinetics."
03/2026: Oral talk (On-Demand) at the APS Global Physics Summit 2026 on "Machine Learning and GPU-Accelerated Modeling of Phase-Ordering Kinetics."
03/2026: Poster presentation (On-Demand) at the APS Global Physics Summit 2026 on "Charge Regulation Effects on Polyelectrolyte-Mediated Nanoparticle Interactions."
05/2023: Poster presentation at the CECAM workshop Mesoscale Modelling of Driven Disordered Materials: From Glasses to Active Matter, on "Study of Hessian Matrix Eigenvalues of the Glassy System Before and After Yield Stress and Its Prediction Using Random Matrix Theory."

FELLOWSHIPS AND AWARDS

2026: Article “Charge Regulation Effect on Nanoparticle Interaction Mediated by a Polyelectrolyte” featured on the front cover of *Macromolecules*, Vol. 59, Issue 11.
2024 – 2027: Senior Research Fellowship (SRF) from the Ministry of Education, India.
2022 – 2024: Junior Research Fellowship (JRF) from the Ministry of Education, India.

SKILLS

Programming Languages

Python Mojo

Simulation Tools/Methods

LAMMPS Finite Difference Method

Machine Learning Frameworks

PyTorch JAX Scikit-Learn

Visualization Tools

OVITO Matplotlib

Additional Skills

SLURM Cluster Git LaTeX

LANGUAGES

English — Fluent

Hindi — Mother tongue